Costo del control a cero y método de Lebeau-Robbiano

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Linear systems of ODEs

$$\dot{y}(t) = Ay(t) + Bu(t) \tag{1}$$

where $y(t) \in \mathbb{R}^n$ is the state, $u(t) \in \mathbb{R}^m$ is the control and $A \in \mathbb{R}^{n\times n}, B \in \mathbb{R}^{n\times m}$ are constant matrices.

Definition (Controllability)

Let $T > 0$. System 1 is said to be controllable in time T when, for any initial state $y_0 \in \mathbb{R}^n$ and any final state $y_T \in \mathbb{R}^n$, there exists $u \in L^p(0, T; \mathbb{R}^m)$ such that the solution to 1 with $y(0) = y_0$ satisfies $y(T) = y_T$.

Cost of control

Idea: Control by increasing finite dimensional subspaces generated by eigenfunctions.

Definition

For $T > 0$, we define the null-control cost as

$$\mathcal{K}^p(T) := \sup_{y_0 \in \mathbb{R}^n,\ |y_0|=1} \inf\{\|u\|_{L^p} \mid u \in L^p(0,T;\mathbb{R}^m),\ y(0) = y_0,\ y(T) = 0\}. \tag{2}$$

Proposition

Let $0 \leq K \leq n-1$ be the minimal integer such that Kalman's criterion is satisfied:

$$\operatorname{rank}[B, AB, A^2B, \ldots, A^K B] = n.$$

Then, there exists $c > 0$ such that

$$\mathcal{K}^p(T) \sim cT^{-K-1+1/p}, \qquad \text{as } T \to 0.$$

Proof of the Proposition pt. 1

Step 1: *Time scaling*. Given $u \in L^p(0, T; \mathbb{R}^m)$, we introduce $v(s) := u(sT)$. From Duhamel's formula at time $t = T$, the condition $y(T) = 0$ is equivalent to

$$F_T(v) := \int_0^1 e^{-s(TA)} Bv(s)ds = -\frac{y_0}{T}. \tag{3}$$

Since $\|u\|_{L^p(0,T)} = T^{1/p}\|v\|_{L^p(0,1)}$, we obtain that

$$\mathcal{K}^p(T) = T^{1/p-1}\tilde{\mathcal{K}}(T), \tag{4}$$

where

$$\tilde{\mathcal{K}}(T) := \sup_{z\in\mathbb{R}^n,\ |z|=1} \text{ínf}\{\|v\|_{L^p} \mid F_T(v) = z\}.$$

Proof of the Proposition pt.2

Step 2: *Decomposition of the state space.* Write $B = (b_1, \dots, b_m)$. Define

$$S_k := \operatorname{span}\{A^j b_l \mid 0 \le j \le k,\ 1 \le l \le m\} \subset \mathbb{R}^n, \qquad S_k = S_{k-1} \oplus E_k, \quad 0 \le k$$

By assumption, $E_0 \oplus \cdots \oplus E_k = \mathbb{R}^n$ and let $P_k : \mathbb{R}^n \to E_k$ be the orthogonal projection on E_k. Then $P_k A^j b_l = 0$ for any $j < k$.
Define

$$\Delta_T(z) := \sum_{k=0}^{K} T^{-k} P_k z, \qquad \Delta_T^{-1}(z) := \sum_{k=0}^{K} T^{k} P_k z.$$

Proof of the Proposition pt.3

Step 3: *Asymptotic expansion of F_T.* Expanding the exponential in 3 we get $\Delta_T F_T = \Lambda + R_T$.

$$\Lambda(v) := \sum_{k=0}^{K} \sum_{l=1}^{m} \mathbb{P}_k(A^k b_l) \int_0^1 \frac{(-s)^k}{k!} v_l(s) ds$$

$$R_T(v) := \sum_{k=0}^{K} \sum_{l=1}^{m} \sum_{j=k+1}^{\infty} \frac{\mathbb{P}_k(A^j b_l)}{j!} T^{j-k} \int_0^1 (-s)^j v_l(s) ds,$$

where $v \in L^p(0,1;\mathbb{R}^m)$. In particular, F_T scales like T^k on each E_k. For $T \leq 1$ we get

$$|R_T(v)| \leq (K+1) \sum_{l=1}^{m} \sum_{j=0}^{+\infty} \frac{\|A\|^j |b_l|}{j!} T \|v_l\|_{L^1} \leq T C_R \|v\|_{L^p}. \tag{5}$$

Proof of the Proposition pt.4

Step 4: $\Lambda : L^p(0,1;\mathbb{R}^m) \to \mathbb{R}^n$ *is surjective.*

We construct a preimage for Λ. Since each E_k is spanned by the $P_k(A^k b_l)$, for each $z \in \mathbb{R}^n$, there exists $\alpha_0, \ldots, \alpha_K \in \mathbb{R}^m$ such that $z = \sum_{k=0}^{K} P_k(A^k B \alpha_k)$.

Moreover, for each $0 \leq k \leq K$ there exists $\varphi_k \in C^\infty([0,1];\mathbb{R})$ such that $\int_0^1 (-s)^j \varphi_k(s) ds = j! \mathbf{1}_{j=k}$ for all $0 \leq j \leq K$. Thus using $v(s) := \sum_{k=0}^{K} \varphi_k(s)\alpha_k$ we get that $\Lambda(v) = z$.

In particular, there exists $U \in \mathcal{L}(\mathbb{R}^n; L^p((0,1);\mathbb{R}^m))$ such that $\Lambda \circ U = \mathrm{Id}_{\mathbb{R}^n}$ and $C_\alpha := \|U\| < \infty$.

Proof of the Proposition pt.5

Step 5: *Definition of the constant.* Since Λ is bounded and $E_K \neq \{0\}$,

$$c := \sup_{|z|=1, z\in E_K} \inf\{\|v\|_{L^p} \mid \Lambda(v) = z\} \in (0, C_\Lambda]. \tag{6}$$

For all $\varepsilon > 0$ and $z \in E_K$, there exists a local inverse $v := V_\varepsilon(z)$ such that $\|V_\varepsilon(z)\| \leq (c + \varepsilon)|z|$ and $\Lambda(V_\varepsilon(z)) = z$.

Proof of the Proposition pt.6

Step 6: *Upper bound on the cost.* For all $\varepsilon > 0$, there exists $T_\varepsilon > 0$ such that, for all $T \in (0, T_\varepsilon]$ and all $z \in \mathbb{R}^n$, there exists $v \in L^p((0,1);\mathbb{R}^m)$ such that $F_T(v) = z$ and $\|v\|_{L^p} \leq (c + 2\varepsilon)T^{-K}|z|$.

For $T, \varepsilon > 0$ and $z \in \mathbb{R}^n$, decompose $z = z_K + z'$ where $(z_K, z') \in E_K \oplus E_K^\perp$. To solve $F_T(v) = z$ we set $v := \sum_{i=0}^{\infty} v_i$ where we define

$$v_0 := V_\varepsilon(\Delta_T(z_K)) + U(\Delta_T(z')), \quad \text{and } v_{i+1} = -U(R_T(v_i)).$$

Then by recursion

$$\|v_0\| \leq (c + \varepsilon)T^{-K}|z_K| + C_\Lambda T^{-K+1}|z'| \leq (c + \varepsilon + C_\Lambda T)T^{-K}|z|.$$

Moreover we obtain a normal convergent series since

$$\|v_{i+1}\| \leq C_\Lambda T C_R \|v_i\| \Longrightarrow \|v\| \leq (c + \varepsilon + TC')T^{-K}|z|.$$

By telescoping sums, $F_T(v) = z$ and choosing T_ε small enough, we obtain $\|v\| \leq (c + 2\varepsilon)T^{-K}|z|$.

Proof of the Proposition pt.7

Step 7: *Lower bound on the cost.* For all $\varepsilon > 0$, there exists $T_\varepsilon > 0$ such that, for all $T \in (0, T_\varepsilon]$, there exists $z \in \mathbb{R}^n$ with $z \neq 0$ such that for all $v \in F_T^{-1}(z)$, one has $\|v\|_{L^p} \geq (c - 2\varepsilon) T^{-K} |z|$.

By cost definition, given $\varepsilon > 0$, there exists $z^* \in E_K$ with $z^* \neq 0$ such that $\Lambda(v) = z^*$ implies $\|v\|_{L^p} \geq (c - \varepsilon)|z^*|$. For $T > 0$, set $z_T := \Delta_T^{-1}(z^*)$ and let v_T such that $F_T(v_T) = z_T$. Thus $(\Lambda + R_T)(v_T) = z^*$.

Let $w_T := U(R_T(v_T))$ so $\Lambda(w_T) = R_T(v_T)$. Then $\Lambda(v_T + w_T) = z^* \in E_K$. Hence $\|v_T + w_T\| \geq (c - \varepsilon)|z^*|$. But since $\|w_T\| \leq C_\Lambda T C_R \|v_T\|$, we have that $(1 + T C_\Lambda C_R)\|v_T\| \geq (c - \varepsilon)|z^*| = (c - \varepsilon) T^{-K} |z_T|$. For $T \in (0, T_\varepsilon]$ where $T_\varepsilon := \varepsilon / (C_\Lambda C_R (c - 2\varepsilon))$, we have found $z_T \in \mathbb{R}^n$ with $z_T \neq 0$ such that any v_T satisfying $F_T(v_T) = z_T$ satisfies $\|v_T\| \geq (c - 2\varepsilon) T^{-K} |z_T|$.

Proof of the Proposition pt.8

Step 8: *Conclusion.* We proved $\tilde{\mathcal{K}} \sim cT^{-K}$. Together with 4 we obtain the result.

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Some definitions

Definition

For $u, v \in L^p(\mathbb{R}_+, U)$, denote by $u \diamond_\tau v \in L^p(\mathbb{R}_+; U)$ their $\tau-$concatenation

$$(u \diamond_\tau v)(s) := \begin{cases} u(s), & s \in [0, \tau), \\ v(s-\tau), & s \geq \tau. \end{cases}$$

Definition

We say that Σ is an abstract linear control system when it is a family $(\Sigma_t)_{t \geq 0}$ of continuous linear maps from $Y \times L^p(\mathbb{R}_+; U)$ to Y such that:

1. for all $y_0 \in Y$ and $u \in L^p(\mathbb{R}_+; U)$, the map $t \mapsto \Sigma_t(y_0, u)$ is continuous on $\mathbb{R}_+$ with $\Sigma_0(y_0, u) = y_0$,
2. for all $t, \tau \geq 0$, $y_0 \in Y$ and $u, v \in L^p(\mathbb{R}_+; U)$,

$$\Sigma_{t+\tau}(y_0, u \diamond_\tau v) = \Sigma_t(\Sigma_\tau(y_0, u), v).$$

$\Sigma_t(y_0, u)$ represents the state $y(t) \in Y$ at time $t \geq 0$ being controlled by $u \in L^p(\mathbb{R}_+; U)$ from an initial condition y_0 governed by a PDE.

Lemma

Let $T > 0$. There exists $M_T > 0$ such that $\forall t \in [0, T]$, $\|\Sigma_t\| \leq M_T$.

A general time iteration

Lebeau-Robbiano strategy is a time-iteration argument which balances the cost of small-time null controllability of eigenprojections and its dissipation rate.

Assumption

There exists a family $(P_k)_{k\in\mathbb{N}}$ *of proyections on* Y *and there exists constantes* $C_1, C_2, c_1, c_2, a, b, m > 0$ *with* $a < b$ *such that, for all* $k \in \mathbb{N}$,

- **Control of low modes.** *For all* $T \in (0,1]$ *and* $y_0 \in Y$, *there exists* $u \in L^p(\mathbb{R}_+; U)$ *such that*

$$P_k \Sigma_T(y_0, u) = 0 \quad \text{and} \quad \|u\|_{L^p} \leq C_1 e^{c_1 k^a} \|y_0\|_Y, \tag{7}$$

- **Dissipation of high modes.** *For all* $T \in (0,1]$ *and* $y_0 \in Y$,

$$P_k y_0 = 0 \quad \Rightarrow \quad \|\Sigma_T(y_0, 0)\|_Y \leq C_2 e^{-c_2 T^m k^b} \|y_0\|_Y. \tag{8}$$

Schematic process

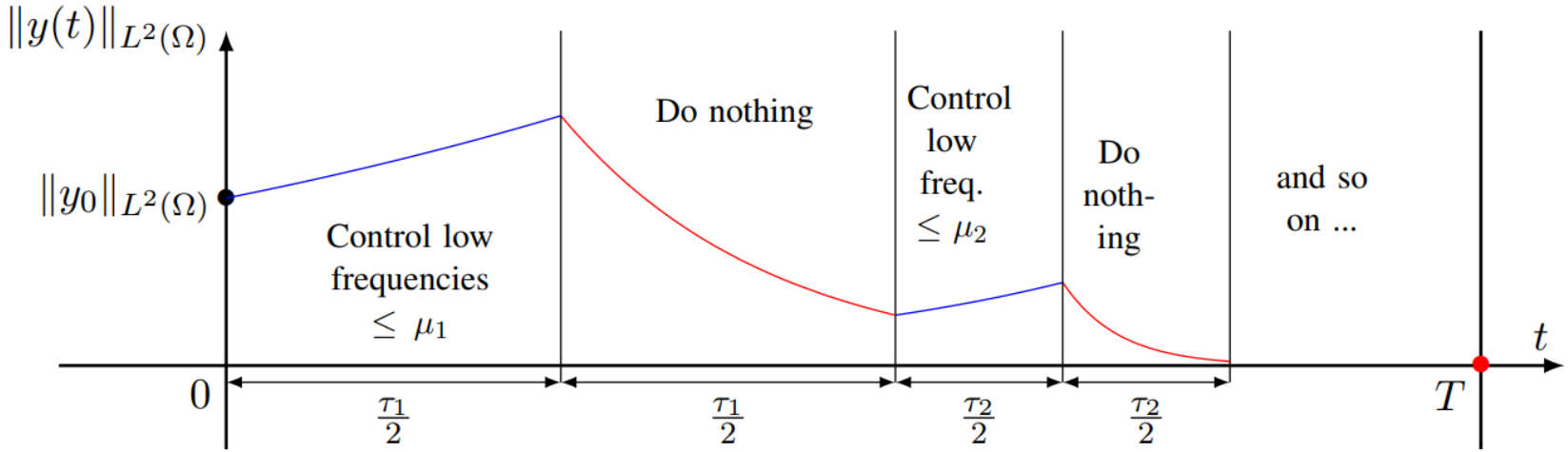


Figure IV.1: The Lebeau-Robbiano method

F. Boyer (2023)
Controllability of linear parabolic equations and systems.

One-step contraction (OSC)

Idea: Partition time interval to apply one of the previous estimates. In each $\tau-$step, control low modes to zero on $[0, \tau/2]$ and let energy dissipate on $[\tau/2, \tau]$. The main time iteration relies on the following one-step contraction result:

Lemma

Suppose that Assumption 2.1 holds. Then there exists $C_3, c_3 > 0$ such that, for all $k \in \mathbb{N}$, for all $\tau \in (0, 1]$ and $y_0 \in Y$, there exists $u \in L^p(\mathbb{R}_+; U)$ such that

$$\|u\|_{L^p} \leq C_1 e^{c_1 k^a} \|y_0\|_Y, \tag{9}$$

$$\|\Sigma_\tau(y_0, u)\|_Y \leq C_3 e^{c_1 k^a} e^{-c_3 \tau^m k^b} \|y_0\|_Y. \tag{10}$$

Proof of the lemma

Assume Assumption 2.1 and use the same constants therein. Let $\tau > 0$, $k \in \mathbb{N}$ and $y_0 \in Y$. Define a control u in $[0, \tau]$ as given in 7 for a time $\tau/2$, extended by 0 on $[\tau/2, \tau]$. Let $y_1 := \Sigma_{\tau/2}(y_0, u)$. By Proposition 5 and Control of Low Modes,

$$\|y_1\|_Y \leq M_1(\|y_0\|_Y + \|u\|_{L^p}) \leq M_1(1 + C_1 e^{c_1 k^a})\|y_0\|_Y.$$

Since $P_k y_1 = 0$, by Dissipation of High Modes implies that

$$\|\Sigma_\tau(y_0, u)\|_Y = \|\Sigma_{\tau/2}(y_1, 0)\| \leq C_2 e^{-c_2(\tau-\tau/2)^m k^b}\|y_1\|_Y,$$

we deduce the claimed estimates with $c_3 = c_2 2^{-m}$ and $C_3 = M_1 C_2(1 + C_1)$.

Theorem

Theorem

Suppose that Assumption 2.1 holds. Then the abstract linear system Σ is small-time null-controllable. More precisely, there exists constants $C_0, c_0 > 0$ such that, for every $T \in (0,1]$ and $y_0 \in Y$, there exists $u \in L^p((0,T);U)$ such that $\Sigma_T(y_0,u) = 0$ and

$$\|u\|_{L^p} \leq C_0 \exp\left(c_0 T^{-\sigma}\right) \|y_0\|_Y, \qquad \sigma := \frac{am}{b-a}. \tag{11}$$

Proof of the Theorem pt. 1

Step 1. *Construction of the time and frequency sequences.* We define a sequence of time intervals and frequency cutoffs. Let $r > 1$ and $q > 1$ be two real numbers to be chosen later. We decompose the interval $[0, T]$ into series of subintervals. Let

$$\alpha := T\left(1 - \frac{1}{r}\right), \quad \text{and} \quad \tau_j := \frac{\alpha}{r^j}, \quad j \in \mathbb{N}. \tag{12}$$

Hence $T = \sum_{j=0}^{\infty} \tau_j$. We define a sequence of time instants $T_0 = 0$ and $T_{j+1} = T_j + \tau_j$. The j-step of our procedure will happen on the interval $[T_j, T_{j+1}]$.
Let $\beta > 0$ be another parameter to be chosen. We define the sequence of frequency cutoffs as

$$k_j := \lfloor \beta q^j \rfloor, \quad \forall j \in \mathbb{N}. \tag{13}$$

Proof of the Theorem pt. 2

Step 2. *The iteration scheme.* We define a sequence of states $(y_j)_{j\in\mathbb{N}}$ by y_0 the given initial estimate and $y_{j+1} := \Sigma_{\tau_j}(y_j, u_j)$ where u_j is the control used on $[T_j, T_{j+1}]$ given by Proposition OSC 6. Hence, by the estimates from OSC 6 we get

$$\|u_j\|_{L^p} \leq C_1 e^{c_1 k_j^a} \|y_j\|_Y, \tag{14}$$

$$\|y_{j+1}\|_Y \leq C_3 e^{-\gamma_j} \|y_j\|_Y, \tag{15}$$

where we introduced the notation $\gamma_j := c_3 \tau_j^m k_j^b - c_1 k_j^a$.

Proof of the Theorem pt. 3

Step 3. *Choice of the parameters* r, q, β. Our goal is to ensure that $\|y_j\|_Y \to 0$ sufficiently fast. For $\beta \geq 1$, by the definition of k_j we know that $\beta q^j/2 \leq k_j \leq \beta q^j$. Recalling τ_j definition, we obtain

$$\gamma_j \geq (q^a)^j \left(2^{-b} c_3 \alpha^m \beta^b (q^{b-a} r^{-m})^j - c_1 \beta^a\right). \tag{16}$$

We let $q := 2^{1/a}$ so that $q^a = 2$ and $r := q^{\frac{b-a}{m}}$, hence $q^{b-a} r^{-n} = 1$. Also, we tune β such that $2^{-b} c_3 \alpha^m \beta^b = 3 c_a \beta^a$, this corresponds to setting

$$\beta := \left(\frac{2^b 3 c_1}{c_3 \alpha^m}\right)^{\frac{1}{b-a}}. \tag{17}$$

With these choices $\beta \to +\infty$ as $T \to 0$ and one has

$$\gamma_j \geq 2 c_1 \beta^a 2^j.$$

Proof of the Theorem pt. 4

Step 4. *Stepwise convergence of the state.* Iterating the construction from step 2, we obtain for any $j \geq 1$,

$$\|y_j\|_Y \leq \exp\left(j \ln C_3 - 2c_1\beta^a(2^j - 1)\right) \|y_0\|_Y. \tag{18}$$

Therefore $\|y_j\|_Y \to 0$ as $j \to \infty$.

Proof of the Theorem pt. 5

Step 5. *Bounding the total control cost.* The overall control u is teh concatenation of the controls u_j on $[T_j, T_{j+1}]$. Using 14 and 18,

$$\begin{aligned}\|u_j\|_{L^p} &\leq C_1 e^{c_1 k_j^a} \|y_j\|_Y \\ &\leq C_1 \exp(c_1 \beta^a 2^j) \exp(j \ln C_3 - 2c_1\beta^a(2^j - 1))\|y_0\|_Y \\ &\leq C_1 e^{c_1\beta^a} \exp(j(\ln C_3 - c_1\beta^a))\|y_0\|_Y \\ &\leq C_1 e^{c_1\beta^a} 2^{-j}\|y_0\|_Y.\end{aligned}$$

Using $2^j - 1 \geq j$ and T small enough so taht $\ln C_3 - c_1\beta^a \leq -\ln 2$. Thus, there exists $T_0 > 0$ such that for all $0 < T \leq T_0$,

$$\|u\|_{L^p((0,T);U)} \leq \sum_{j=0}^{\infty} \|u_j\|_{L^p((T_j,T_{j+1});U)} \leq 2C_1 e^{c_1\beta^a}\|y_0\|_Y. \tag{19}$$

Recalling 12 and 17, this entails the cost estimate 11 since $\beta^a \approx (\alpha^{-\frac{m}{b-a}})^a \approx T^{-\sigma}$.

Proof of the Theorem pt. 6

Step 6. *Conclusion.* Since Σ is an abstract linear control system, the associated solution $y(t) := \Sigma_t(y_0, u)$ belongs to $C^0([0, T]; Y)$. By the composition property of the $\tau-$concatenation,$y(T_j) = y_j$. We proved that $y_j \to 0$. Since $T_j \to T$, by continuity we conclude that $\Sigma_T(y_0, u) = y(T) = 0$.

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Lebeau-Robbiano strategy

Given an spatial operator A which admits an eigen-decomposition with projections $(P_k)_{k\in\mathbb{N}}$,

1. Prove an energy estimate dissipation. (Control of low modes).
2. Prove a dissipation property for high modes. (Spectral inequality!)
3. Do we know if P_k and $\mathbb{T}_t$ commutate?
4. With the previous estimates, our Assumption 1 holds. Apply Theorem 7 to achieve null-controllability.

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Example: interior control of the heat equation on [0,1].

$$\begin{cases} y_t - y_{xx} = u1_\omega, & (0,T)\times(0,1), \\ y(t,0) = y(t,1) = 0, & (0,T), \\ y(0,x) = y_0(x), & (0,1). \end{cases} \tag{20}$$

$y_0 \in L^2(0,1)$, $u \in L^2((0,T)\times\omega)$ and $\omega \subset (0,1)$. The system is well-posed in

$$\mathcal{Y} := C^0([0,T]; L^2(0,1)) \cap L^2((0,T); H_0^1(0,1)) \cap H^1((0,T); H^{-1}(0,1)),$$

Moreover, there exists a constant $C > 0$ such that for every T, f, y_0,

$$\|y\|_{C^0L^2} + \|y\|_{L^2H^1} + \|y\|_{H^1H^{-1}} \leq C(\|y_0\|_{L^2} + \|f\|_{L^2H^{-1}}).$$

Energy dissipation and spectral inequality

$A = -\partial_{xx}$ with Dirichlet conditions admits the eigen-decomposition

$$\varphi_j(x) := \sqrt{2}\sin(j\pi x), \qquad \lambda_j = (j\pi)^2.$$

If $\langle y_0, \varphi_j \rangle = 0$ for $1 \leq j \leq k$, one has

$$\|y(t)\|_{L^2} \leq e^{-(k\pi)^2 t}\|y_0\|_{L^2}. \tag{21}$$

We assume the next spectral inequality due to Turán,

Proposition

Let $\omega \subset [0,1]$ be a measurable set with $|\omega| > 0$. there exists $C_\omega > 0$ such that for any $n \geq 1$ and $a_1, \ldots, a_n \in \mathbb{R}$,

$$\|f\|_{L^2(0,1)} \leq C_\omega^n \|f\|_{L^2(\omega)}, \quad f(x) := \sum_{j=1}^{n} a_j \sin(j\pi x).$$

Small-time null controllability

Theorem

Assume that $|\omega| > 0$. *Previous system is small-time null controllable and there exists* $C > 0$ *such that, for all* $T > 0$ *and* $y_0 \in L^2(0,1)$, *there exists* $u \in L^2((0,T) \times \omega)$ *such that the associated solution to* (20) *satisfies* $y(T) = 0$ *and* $\|u\|_{L^2} \leq ce^{C/T}\|y_0\|_{L^2}$.

Proof: Denote by P_k the projection on $\text{span}\{\varphi_1, \ldots, \varphi_k\}$ in $L^2(0,1)$. Energy dissipation of high modes holds with $b = 2$ and $m = 1$. Since P_k commutes with $\mathbb{T}_t$ and $B \in \mathcal{L}(L^2(\omega); L^2(0,1))$ is the extension by 0, so B^* is the restriction to ω. Hence by the spectral inequality, for any $y \in L^2(0,1)$,

$$\|P_k y\|_Y \leq e^{c_\omega k}\|B^* P_k y\|_U.$$

Thus we can apply the next Proposition

$$P_k \Sigma_T(y_0, u) = 0, \quad \text{and} \quad \|u\|_{L^p} \leq T^{1/p-1} M_1 C_1 e^{c_1 k^a}\|y_0\|_Y,$$

which implies that the control cost estimate holds with $a = 1$.

End of the proof

Thus Proposition 2.1 holds and Theorem 7 applies.
Since $p = 2$, the cost estimate from previous Proposition leads to a singular $T^{-1/2}$ prefactor, which is an absorbable weak degeneracy with the same iteration.

What's next

1. We used the next result which applies when P_k and $\mathbb{T}_t$ commutes

Proposition

Assuming that a Spectral Inequality holds, if the projections P_k commute with the dynamics, then, for all $k \in \mathbb{N}$, $T \in (0,1]$ and $y_0 \in Y$, there exists $u \in L^p((0,T);U)$ such that

$$P_k \Sigma_T(y_0, u) = 0, \quad \text{and} \quad \|u\|_{L^p} \leq T^{1/p-1} M_1 C_1 e^{c_1 k^a} \|y_0\|_Y, \tag{22}$$

where M_1 is the constant from Lemma 5.

2. There is another result which applies when we do not have the commutation property

Proposition

Assume that there exist $C_1, C_2, c_1, c_2, a, b, m > 0$ *with* $a < b$ *such that, for all* $k \in \mathbb{N}$, *the spectral inequality applies and the dissipation condition 8 hold. Then there exists* $C_3, c_3 > 0$ *such that, for all* $k \in \mathbb{N}$, *for all* $k \in \mathbb{N}$, *for all* $\tau \in (0, 1]$ *and* $y_0 \in Y$, *there exists* $u \in L^p(\mathbb{R}_+; U)$ *with*

$$\|u\|_{L^p} \leq 2\tau^{1/p-1} M_1 C_1 e^{c_1 k^a} \|y_0\|_Y, \tag{23}$$

$$\|\Sigma_\tau(y_0, u)\|_Y \leq C_3 e^{c_1 k^a} e^{-c_3 \tau^m k^b} \|y_0\|_Y, \tag{24}$$

where M_1 *is the constant from Lemma 5.*

3. Next session: Demonstrate the previous results.
4. Then we can show another examples applying previous results.